1 From the electron-phonon interaction to the BCS ground state

1.1 Overview

The road from deriving the Hamiltonian of the electron–phonon interaction to the BCS Hamiltonian, the physics it contains and finally its ground state is quite long and necessitates many calculations. Hence, this note only gives a concise summary and then refers to the literature for more information. I mostly follow *Solid State Theory 2* by Gerd Czycholl; equation numbers of the form (C 5.7) refer to that book. The Hamiltonian used as a starting point is derived by Czycholl in Sec. 7.1 of his first book and also by Bruus and Flensberg in Ch. 3 of *Many-Body Quantum Theory in Condensed Matter Physics*. If you are simply interested in the BCS ground state, you can directly skip to section 1.1.4. If you want the details, refer to section 1.2.

1.1.1 Attractive electron–electron interaction through the electron-phonon mechanism

Intuitive picture/motivation.

An electron e_1 moving through the lattice distorts it, i.e. it drags the positive ions towards itself and leaves behind a region of enhanced positive charge. A second electron e_2 then feels an attraction to e_1 through this phonon excitation caused by e_1. The result is an effective electron–electron interaction that is attractive.

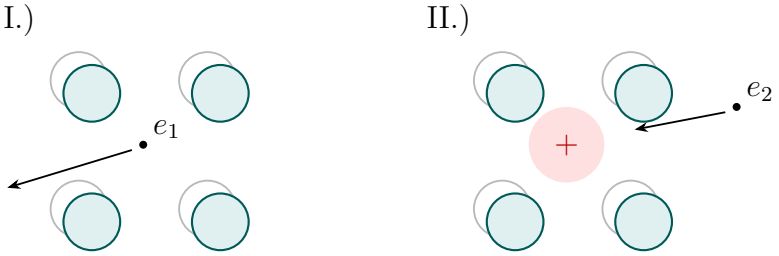


Figure 1: The lattice distortion caused by e_1 (I.) outlives its passage and attracts a second electron e_2 (II.).

Starting point.

The starting point for all our derivations is the Fröhlich Hamiltonian, derived in Czycholl Band 1, Sec. 7.1 (7.13)–(7.15) or Bruus and Flensberg, Sec. 3.8, (C 5.7):

$$H = \sum_{\boldsymbol{k},\sigma} \varepsilon_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k},\sigma} c_{\boldsymbol{k},\sigma} + \sum_{\boldsymbol{q}} \hbar\omega_{\boldsymbol{q}}\, b^\dagger_{\boldsymbol{q}} b_{\boldsymbol{q}} + \sum_{\boldsymbol{k},\boldsymbol{q},\sigma} \Big[M(\boldsymbol{q})\, c^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma} c_{\boldsymbol{k},\sigma} b_{\boldsymbol{q}} + M(-\boldsymbol{q})\, c^\dagger_{\boldsymbol{k}-\boldsymbol{q},\sigma} c_{\boldsymbol{k},\sigma} b^\dagger_{\boldsymbol{q}} \Big], \quad (1)$$

with $\omega_{-\boldsymbol{q}} = \omega_{\boldsymbol{q}}$ and $M(-\boldsymbol{q}) = M(\boldsymbol{q})$.

Canonical transformation.

Use a canonical transformation[1] $H \rightsquigarrow H_T = e^{-iS} H e^{iS}$ to find a purely electronic Hamiltonian that describes this interaction, i.e. to get rid of processes that change the number of phonons. At one point one simply restricts the Hamiltonian not to mix states with different phonon configurations.

The canonical transformation yields $H_T \cong H_0 + H_{\text{int}}$ with the interaction (C 5.25)

$$H_{\text{int}} = -\frac{1}{2} \sum_{\sigma,\sigma'} \sum_{\boldsymbol{k},\boldsymbol{k}',\boldsymbol{q}} V_{\boldsymbol{k},\boldsymbol{k}',\boldsymbol{q}}\; c^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma}\, c^\dagger_{\boldsymbol{k}'-\boldsymbol{q},\sigma'}\, c_{\boldsymbol{k}',\sigma'}\, c_{\boldsymbol{k},\sigma}, \tag{2}$$

which is attractive if $|\varepsilon_{\boldsymbol{k}} - \varepsilon_{\boldsymbol{k}-\boldsymbol{q}}| \leq \hbar\omega_{\boldsymbol{q}} \leq \hbar\omega_D$ in $V_{\boldsymbol{k},\boldsymbol{k}',\boldsymbol{q}}$, where ω_D is the Debye frequency, and can be represented as in fig. 2.

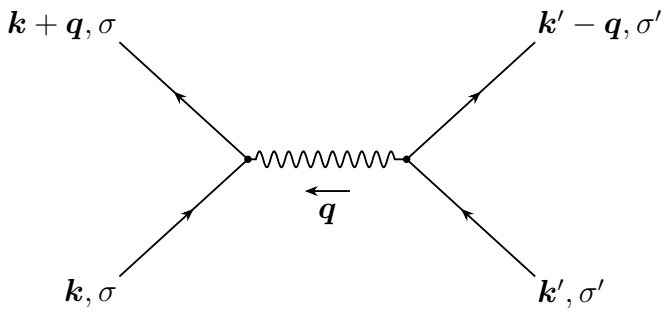


Figure 2: Vertex corresponding to eq. (2): two electrons exchange a phonon of momentum $\boldsymbol{q}$.

Since we know that the momentum transfer is $\boldsymbol{q}$, we know that $(\boldsymbol{k},\sigma) \to (\boldsymbol{k}'-\boldsymbol{q},\sigma')$ is not a valid fermion line.

1.1.2 Cooper pairs

In order to examine how this attractive interaction might change the many-body ground state, one looks at a highly simplified model:

- a system of free, non-interacting electrons, i.e. the Fermi sea (C 5.27)

$$|\phi_0\rangle = \prod_{\varepsilon_{\boldsymbol{k}} \leq \varepsilon_F,\,\sigma} c^\dagger_{\boldsymbol{k},\sigma} |0\rangle, \tag{3}$$

- the effective interaction is only present for two electrons that are placed right outside of the Fermi sea with opposite momenta and opposite spins (Cooper pair, singlet pairing);
- the interaction is $V = \text{const.}$ if $|\varepsilon_{\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{\boldsymbol{k}}| \leq \hbar\omega_D$ and zero otherwise.

[1] See **??** for more general information on similarity transformations.

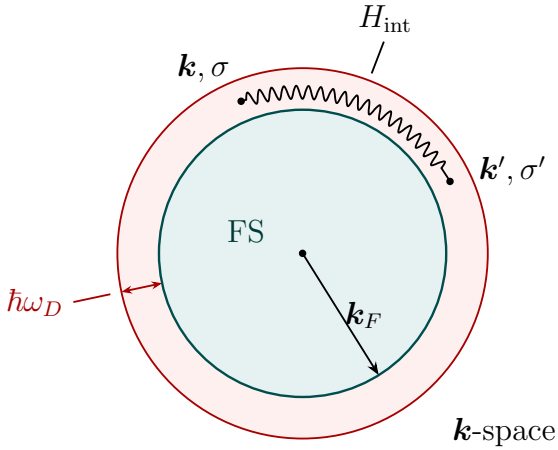


Figure 3: Only states in a shell of width $\hbar\omega_D$ around the Fermi surface take part in the effective interaction.

Here "right outside the Fermi sea" means that the energies are, cf. (C 5.27),

$$E_F \leq \varepsilon_{\boldsymbol{k}}, \;\; \varepsilon_{-\boldsymbol{k}} \leq E_F + \hbar\omega_D. \tag{4}$$

The interaction thus reads (C 5.28)

$$H_{\text{int}} = -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{\boldsymbol{k},\boldsymbol{q}} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma}\, c^\dagger_{-\boldsymbol{k}-\boldsymbol{q},\sigma'}\, c_{-\boldsymbol{k},\sigma'}\, c_{\boldsymbol{k},\sigma}, \qquad \text{with } |\varepsilon_{\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{\boldsymbol{k}}| \leq \hbar\omega_D. \tag{5}$$

One solves this model by making the variational ansatz (C 5.26)

$$|\psi_{\sigma,\sigma'}\rangle = \sum_{\boldsymbol{k}} a_{\sigma,\sigma'}(\boldsymbol{k})\, c^\dagger_{\boldsymbol{k},\sigma}\, c^\dagger_{-\boldsymbol{k},\sigma'}\, |\phi_0\rangle \tag{6}$$

and minimizing the energy $E = \langle\psi_{\sigma,\sigma'}|H|\psi_{\sigma,\sigma'}\rangle$ with the constraint that $\langle\psi_{\sigma,\sigma'}|\psi_{\sigma,\sigma'}\rangle \stackrel{!}{=} 1$, which gets us an upper bound for the energy of the bound state. One finds that (C 5.35)

$$\Delta E = 2E_F - E_{\text{bs}} \cong -2\hbar\omega_{\boldsymbol{q}}\, e^{-2/(V\rho_0)}, \tag{7}$$

which shows that a bound state (Cooper pair) of the two electrons is energetically favoured compared to the free, non-interacting state. This holds for all electrons close to the Fermi edge and therefore implies an instability of the Fermi sea.

1.1.3 BCS theory

To solve the Hamiltonian and extract the relevant mechanism from it, it is simplified as follows: Only include interactions between electrons with opposite momenta and opposite spins, and neglect the $\boldsymbol{q}$-dependence. This yields (C 5.38)

$$H = \sum_{\boldsymbol{k},\sigma} \varepsilon_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\sigma} c_{\boldsymbol{k}\sigma} - V \sum_{\boldsymbol{k},\boldsymbol{k}'} c^\dagger_{\boldsymbol{k}'\uparrow} c^\dagger_{-\boldsymbol{k}'\downarrow} c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow}, \tag{8}$$

with again $|\varepsilon_{\boldsymbol{k}} - \varepsilon_{\boldsymbol{k}'}| \leq \hbar\omega_D$ for the interaction term. Finally, apply mean-field theory to the pair creation/destruction operators, (C 5.41) and (C 5.42):

$$H_{\mathrm{MF}} = \sum_{\boldsymbol{k}} \begin{pmatrix} c_{\boldsymbol{k}\uparrow}^\dagger & c_{-\boldsymbol{k}\downarrow} \end{pmatrix} \begin{pmatrix} \varepsilon_{\boldsymbol{k}} & -\Delta \\ -\Delta^* & -\varepsilon_{\boldsymbol{k}} \end{pmatrix} \begin{pmatrix} c_{\boldsymbol{k}\uparrow} \\ c_{-\boldsymbol{k}\downarrow}^\dagger \end{pmatrix} \pm \text{ const.}, \tag{9}$$

$$\Delta = V \sum_{\boldsymbol{k}} \langle c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow} \rangle \qquad (\text{possibly } \varepsilon_{\boldsymbol{k}} \rightsquigarrow \varepsilon_{\boldsymbol{k}} - \mu). \tag{10}$$

Bogoliubov transformation.

Introduce a unitary transformation that diagonalizes

$$B = \begin{pmatrix} \varepsilon_{\boldsymbol{k}} & -\Delta \\ -\Delta^* & -\varepsilon_{\boldsymbol{k}} \end{pmatrix}$$

and thus leads to an H_{MF} that only includes terms of the form $\alpha_{\boldsymbol{k}}^\dagger \alpha_{\boldsymbol{k}}$ and $\beta_{\boldsymbol{k}}^\dagger \beta_{\boldsymbol{k}}$:

$$H_{\mathrm{MF}} = \sum_{\boldsymbol{k}} \underbrace{\begin{pmatrix} c_{\boldsymbol{k}\uparrow}^\dagger & c_{-\boldsymbol{k}\downarrow} \end{pmatrix} U_{\boldsymbol{k}}}_{\text{new operators}} \underbrace{U_{\boldsymbol{k}}^\dagger \begin{pmatrix} \varepsilon_{\boldsymbol{k}} & -\Delta \\ -\Delta^* & -\varepsilon_{\boldsymbol{k}} \end{pmatrix} U_{\boldsymbol{k}}}_{\text{diagonal matrix}} \underbrace{U_{\boldsymbol{k}}^\dagger \begin{pmatrix} c_{\boldsymbol{k}\uparrow} \\ c_{-\boldsymbol{k}\downarrow}^\dagger \end{pmatrix}}_{(\text{new operators})^\dagger}. \tag{11}$$

These new operators will be fermionic, since U is unitary, so that $\langle c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow} \rangle \rightsquigarrow f(\alpha_{\boldsymbol{k}}, \beta_{\boldsymbol{k}}, \alpha_{\boldsymbol{k}}^\dagger, \beta_{\boldsymbol{k}}^\dagger)$ and one can then use $\langle \alpha_{\boldsymbol{k}}^\dagger \alpha_{\boldsymbol{k}} \rangle = f(E_F)$ and $\langle \beta_{\boldsymbol{k}} \alpha_{\boldsymbol{k}} \rangle = 0$ etc. The elements of U can be determined by diagonalizing B and requiring that the eigenvectors are normalized to one.

Result.

This yields

$$H_{\mathrm{eff}} = \sum_{\boldsymbol{k}} E_{\boldsymbol{k}} \big(\alpha_{\boldsymbol{k}}^\dagger \alpha_{\boldsymbol{k}} + \beta_{\boldsymbol{k}}^\dagger \beta_{\boldsymbol{k}} \big) + \text{const.}, \qquad (\text{C } 5.46) \tag{12}$$

$$E_{\boldsymbol{k}} = \big(\varepsilon_{\boldsymbol{k}}^2 + |\Delta|^2 \big)^{1/2} \ \geq \ 0, \qquad (\text{C } 5.47) \tag{13}$$

and for Δ one arrives at the BCS self-consistency equation (C 5.50)

$$\Delta = \frac{V\Delta}{2} \sum_{\boldsymbol{k}} \frac{1}{E_{\boldsymbol{k}}} \tanh\!\left(\frac{\beta E_{\boldsymbol{k}}}{2} \right). \tag{14}$$

Obviously $\Delta = 0$ is a solution. For $\Delta \neq 0$ one obtains (C 5.51)

$$1 = \frac{V}{2} \sum_{\boldsymbol{k}, |\varepsilon_{\boldsymbol{k}}| < \hbar\omega_D} \big(\varepsilon_{\boldsymbol{k}}^2 + |\Delta|^2 \big)^{-1/2} \tanh\!\left(\frac{\beta \big(\varepsilon_{\boldsymbol{k}}^2 + |\Delta|^2 \big)^{1/2}}{2} \right). \tag{15}$$

Δ as an order parameter.

Actually, Δ is an order parameter for the superconducting phase, because of the following:

- For a normal metal $\langle c_{\boldsymbol{k}\uparrow} c_{-\boldsymbol{k}\downarrow}\rangle = 0 \;\forall\, \boldsymbol{k}$ and thus $\Delta = V \sum_{\boldsymbol{k}} \langle c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow}\rangle = 0$.
- But if $\Delta \neq 0$, it means that we are not in a metallic phase and that there are Cooper pairs present.

1.1.4 The BCS ground state

We have $H = \sum_{\boldsymbol{k}} E_{\boldsymbol{k}} (\alpha^\dagger_{\boldsymbol{k}} \alpha_{\boldsymbol{k}} + \beta^\dagger_{\boldsymbol{k}} \beta_{\boldsymbol{k}})$ with $E_{\boldsymbol{k}} \geq 0$, cf. eq. (12), where, following (C 5.43),

$$\alpha_{\boldsymbol{k}} = u_{\boldsymbol{k}} c_{\boldsymbol{k}\uparrow} - v_{\boldsymbol{k}} c^\dagger_{-\boldsymbol{k}\downarrow} \qquad \text{and} \qquad \beta_{\boldsymbol{k}} = u_{\boldsymbol{k}} c_{-\boldsymbol{k}\downarrow} + v_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}\uparrow}. \tag{16}$$

Since $\alpha^{(\dagger)}_{\boldsymbol{k}}$ and $\beta^{(\dagger)}_{\boldsymbol{k}}$ are fermionic, the eigenvalues of $\alpha^\dagger_{\boldsymbol{k}} \alpha_{\boldsymbol{k}}$ (and $\beta^\dagger_{\boldsymbol{k}} \beta_{\boldsymbol{k}}$) are $0, 1$. Hence the ground state of H does not contain any α or β quasiparticles, i.e.

$$\alpha_{\boldsymbol{k}} |\mathrm{BCS}\rangle = \beta_{\boldsymbol{k}} |\mathrm{BCS}\rangle = 0. \tag{17}$$

This is satisfied by

$$|\mathrm{BCS}\rangle = \prod_{\boldsymbol{k}} \left(u_{\boldsymbol{k}} + v_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow}\right)|0\rangle = \bigotimes_{\boldsymbol{k}} \left(u_{\boldsymbol{k}} + v_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow}\right)|0\rangle_{\boldsymbol{k}}. \tag{18}$$

1.2 Details

1.2.1 Preface

These are supplementary notes to *Solid State Theory, Volume 2* by Gerd Czycholl. Since the derivations there are quite lengthy, only those steps are presented here that clarify what is done in the corresponding chapters. Numbers in parentheses such as (5.12) refer to equations of that book.

1.2.2 Attractive electron-electron interaction through the electron-phonon mechanism (Czycholl 5.2)

Two small observations.

- (5.12): use that S is linear in H_1 to discard $[[H_1, S], S]$.
- (5.15): use that $\langle m|H_1|m\rangle = 0$, since H_1 changes the number of phonons and thus $H_1|m\rangle \perp |m\rangle$.

The transformed Hamiltonian.

Starting from (5.13) and inserting resolutions of the identity,

$$H_T = H_0 + \frac{i}{2}\big(H_1 S - S H_1\big) \tag{19}$$

$$= H_0 + \frac{i}{2}\sum_{\tilde{n},n,m}\Big(|\tilde{n}\rangle\langle\tilde{n}|H_1|m\rangle\langle m|S|n\rangle\langle n| - |\tilde{n}\rangle\langle\tilde{n}|S|m\rangle\langle m|H_1|n\rangle\langle n|\Big) \tag{20}$$

$$\overset{(5.15)}{=} H_0 - \frac{1}{2}\sum_{\tilde{n},n,m}\Big(|\tilde{n}\rangle\langle\tilde{n}|H_1|m\rangle\langle m|H_1|n\rangle\langle n|\frac{1}{E_m - E_n} - |\tilde{n}\rangle\langle\tilde{n}|H_1|m\rangle\langle m|H_1|n\rangle\langle n|\frac{1}{E_{\tilde{n}} - E_m}\Big), \tag{21}$$

where the matrix elements $\langle\tilde{n}|H_1|m\rangle$ and $\langle m|H_1|n\rangle$ are scalars and can be pulled out of the operator structure, which yields (5.16).

Restriction to a fixed phonon configuration.

Since we want to derive a Hamiltonian for the electron-electron interaction only, we restrict $|\tilde{n}\rangle$ and $|n\rangle$ in (5.16) to states with the same phonon configuration, such that H_T does not mix states with different phonon configurations.

Let $|n\rangle$ be some many-particle state and assume that it contains one electron in the state $|k,\sigma\rangle$ and one in $|k',\sigma'\rangle$. The two elementary processes are

$$H_{1,1}|k,\sigma\rangle = \sqrt{n_q}\,|k+q,\sigma,n_q-1\rangle, \qquad \text{by destroying a phonon,} \tag{$*_1$}$$

$$H_{1,2}|k,\sigma\rangle = \sqrt{n_q+1}\,|k-q,\sigma,n_q+1\rangle, \qquad \text{by creating a phonon.} \tag{$*_2$}$$

Since $|\tilde{n}\rangle$ and $|n\rangle$ are restricted to states with the same phonon number, the created (destroyed) phonon with momentum q has to be destroyed (created) again:

$$H_{1,1}\sqrt{n_q}\,|k+q,\sigma,n_q-1\rangle = n_q\,|k+q,\sigma;\,k'-q,\sigma';\,n_q\rangle, \tag{22}$$

$$H_{1,1}\sqrt{n_q+1}\,|k-q,\sigma,n_q+1\rangle = (n_q+1)\,|k-q,\sigma;\,k'+q,\sigma';\,n_q\rangle. \tag{23}$$

All other terms vanish, since their phonon configurations differ.

Energy denominators.

Therefore we have

$$(*_1): \quad E_m - E_n = \varepsilon_{k+q} - \varepsilon_k - \hbar\omega_q, \qquad E_{\tilde{n}} - E_m = \varepsilon_{k'-q} - \varepsilon_{k'} + \hbar\omega_q, \qquad (5.19) \tag{24}$$

$$(*_2): \quad E_m - E_n = \varepsilon_{k-q} - \varepsilon_k + \hbar\omega_q, \qquad E_{\tilde{n}} - E_m = \varepsilon_{k'+q} - \varepsilon_{k'} - \hbar\omega_q. \qquad (5.20) \tag{25}$$

Inserting this into (5.16) gives

$$H_T = -\frac{1}{2}\sum_{n,\tilde{n}}|\tilde{n}\rangle\langle n|\sum_m\sum_{\substack{k,k',q\\ \sigma,\sigma'}}\Big(\langle\tilde{n}|M(-q)c^\dagger_{k'-q,\sigma'}c_{k',\sigma'}b^\dagger_q|m\rangle\langle m|M(q)c^\dagger_{k+q,\sigma}c_{k,\sigma}b_q|n\rangle \times\Big(\frac{1}{\varepsilon_{k+q}-\hbar\omega_q-\varepsilon_k}-\frac{1}{\varepsilon_{k'-q}-\varepsilon_{k'}+\hbar\omega_q}\Big) + \langle\tilde{n}|M(q)c^\dagger_{k'+q,\sigma'}c_{k',\sigma'}b_q|m\rangle\langle m|M(-q)c^\dagger_{k-q,\sigma}c_{k,\sigma}b^\dagger_q|n\rangle \times\Big(\frac{1}{\varepsilon_{k-q}-\varepsilon_k+\hbar\omega_q}-\frac{1}{\varepsilon_{k'+q}-\varepsilon_{k'}-\hbar\omega_q}\Big)\Big), \tag{26}$$

where the first summand is the $(*_1)$-process and the second one the $(*_2)$-process.

Using $M^*(q) = M(-q)$ and that all the above terms are independent of n, m, and $\tilde{n}$ yields (5.21). Using furthermore that

$$b^\dagger_q b_q|\dots n_q\dots\rangle = n_q|\dots n_q\dots\rangle, \qquad b_q b^\dagger_q|\dots n_q\dots\rangle = (n_q+1)|\dots n_q\dots\rangle \tag{27}$$

results in (5.21), i.e. $\hat{H}_T(b_q, b^\dagger_q, \dots) \rightsquigarrow \hat{H}_T(n_q, \dots)$ in the phonon number basis.

Combining the two processes.

We have

$$\frac{1}{\varepsilon_{k+q}-\hbar\omega_q-\varepsilon_k} = \frac{(\varepsilon_{k+q}-\varepsilon_k)+\hbar\omega_q}{(\varepsilon_{k+q}-\varepsilon_k)^2-\hbar^2\omega_q^2} \tag{28}$$

and, relabelling $k \leftrightarrow k'$ and $q \to -q$,

$$\sum_{k,k',q}\frac{1}{\varepsilon_{k'-q}-\varepsilon_{k'}+\hbar\omega_q}c^\dagger_{k+q}c_k c^\dagger_{k'-q}c_{k'} = \sum_{k,k',q}\frac{(\varepsilon_{k+q}-\varepsilon_k)-\hbar\omega_q}{(\varepsilon_{k+q}-\varepsilon_k)^2-\hbar^2\omega_q^2}c^\dagger_{k'-q}c_{k'}c^\dagger_{k+q}c_k, \tag{29}$$

where

$$\sum_{k,k',q}c^\dagger_{k'-q}c_{k'}c^\dagger_{k+q}c_k = \sum_{k,k',q}c^\dagger_{k'-q}\big(\delta_{k',k+q}-c^\dagger_{k+q}c_{k'}\big)c_k \tag{30}$$

$$= \sum_{k,q}c^\dagger_k c_k - \sum_{k,k',q}c^\dagger_{k+q}c^\dagger_{k'-q}c_k c_{k'} \tag{31}$$

$$= \sum_{k,q}n_k - \sum_{k,k',q}c^\dagger_{k+q}\big(\delta_{k,k'-q}-c_k c^\dagger_{k'-q}\big)c_{k'} \tag{32}$$

$$= \underbrace{\sum_{k,q}n_k - \sum_{k,q}c^\dagger_{k'}c_{k'}}_{=0} + \sum_{k,k',q}c^\dagger_{k+q}c_k c^\dagger_{k'-q}c_{k'} \tag{33}$$

$$= \sum_{k,k',q}c^\dagger_{k+q}c_k c^\dagger_{k'-q}c_{k'}. \tag{34}$$

We know (Bruus & Flensberg, Sec. 3.8) that $\omega_{-q} = \omega_q$ and $M(-q) = M(q)$. Furthermore, since we are only interested in the electronic process, we can assume that there is no net phononic motion within the system and thus $n_q = n_{-q}$. Hence the first term of (5.21) equals the first term of (5.22), analogously for the second term, and similarly (5.22) $\rightsquigarrow$ (5.23).

1.2.3 Cooper pairs (Czycholl 5.3)

Setup.

Simplify: look at the Fermi sea plus two electrons just above the Fermi energy, and only let them interact via H_T. The variational ansatz (5.26), for vanishing total pair momentum $K = k + k' \stackrel{!}{=} 0$, is

$$|\psi_{\sigma,\sigma'}\rangle = \sum_k a_{\sigma,\sigma'}(k)\, c^\dagger_{k,\sigma} c^\dagger_{-k,\sigma'} |\phi_0\rangle, \tag{35}$$

with the sum over k such that $E_F \le \varepsilon_k, \varepsilon_{k'} \le E_F + \hbar\omega_D$, cf. (5.27). The simplified interaction acts only outside the Fermi sea $(*)$, otherwise $V = 0$:

$$H_T = -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q} c^\dagger_{k+q,\sigma} c^\dagger_{-k-q,\sigma'} c_{-k,\sigma'} c_{k,\sigma}. \qquad (5.28) \tag{36}$$

Action on the ansatz.

$$H_T|\psi_{\sigma_1\sigma_2}\rangle = -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q,k'} c^\dagger_{k+q,\sigma} c^\dagger_{-k-q,\sigma'} c_{-k,\sigma'} c_{k,\sigma}\, a_{\sigma_1\sigma_2}(k')\, c^\dagger_{k',\sigma_1} c^\dagger_{-k',\sigma_2} |\phi_0\rangle \tag{37}$$

$$= -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q,k'} a_{\sigma_1\sigma_2}(k')\, c^\dagger_{k+q,\sigma} c^\dagger_{-k-q,\sigma'} c_{-k,\sigma'} c_{k,\sigma}\, |k', \sigma_1; -k', \sigma_2\rangle \tag{38}$$

$$= -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q,k'} a_{\sigma_1\sigma_2}(k') \Big(\delta_{k,k'}\delta_{\sigma,\sigma_1}\delta_{\sigma',\sigma_2} |k+q, \sigma_1; -k-q, \sigma_2\rangle + \delta_{k,-k'}\delta_{\sigma,\sigma_2}\delta_{\sigma',\sigma_1} |-k-q, \sigma_1; k+q, \sigma_2\rangle \Big) \tag{39}$$

$$= -\frac{V}{2} \sum_{k,q} \Big(a_{\sigma_1\sigma_2}(k)\, |k+q, \sigma_1; -k-q, \sigma_2\rangle + a_{\sigma_1\sigma_2}(-k)\, |-k-q, \sigma_1; k+q, \sigma_2\rangle \Big), \tag{40}$$

where the projection onto $|k', \sigma_1; -k', \sigma_2\rangle$ is allowed because, by $(*)$, only states outside the Fermi sea matter for the interaction. Taking the expectation value,

$$\langle\psi_{\sigma_1\sigma_2}|H_T|\psi_{\sigma_1\sigma_2}\rangle = -\frac{V}{2} \sum_{k,q,k'} a^*_{\sigma_1\sigma_2}(k') \Big\{ a_{\sigma_1\sigma_2}(k)\, \langle k', \sigma_1; -k', \sigma_2 | k+q, \sigma_1; -k-q, \sigma_2\rangle + a_{\sigma_1\sigma_2}(-k)\, \langle k', \sigma_1; -k', \sigma_2 | -k-q, \sigma_1; k+q, \sigma_2\rangle \Big\} \tag{41}$$

$$= -\frac{V}{2} \sum_{k,q,k'} a^*_{\sigma_1\sigma_2}(k') \Big\{ a_{\sigma_1\sigma_2}(k)\, \delta_{k',k+q} + a_{\sigma_1\sigma_2}(-k)\, \delta_{k',-k-q} \Big\} \tag{42}$$

$$= -\frac{V}{2} \sum_{k,q} \Big(a^*_{\sigma_1\sigma_2}(k+q) a_{\sigma_1\sigma_2}(k) + a^*_{\sigma_1\sigma_2}(-k-q) a_{\sigma_1\sigma_2}(-k) \Big) \tag{43}$$

$$= -V \sum_{k,q} a^*_{\sigma_1\sigma_2}(k+q) a_{\sigma_1\sigma_2}(k), \tag{44}$$

where in the last step $k \to -k$, $q \to -q$ was substituted in the second term.

The kinetic part.

Furthermore, by (5.17),

$$H_0 = \sum_{k,\sigma} \varepsilon_k n_{k,\sigma} + \sum_q \hbar\omega_q b_q^\dagger b_q. \tag{45}$$

Since the ansatz $|\psi_{\sigma\sigma'}\rangle$ is purely electronic, $\sum_q \hbar\omega_q b_q^\dagger b_q$ will not contribute. Splitting the electronic sum into states inside and outside the Fermi sea,

$$\begin{aligned}
\langle\psi_{\sigma,\sigma'}|H_0|\psi_{\sigma,\sigma'}\rangle &= \langle\psi_{\sigma\sigma'}|\Big(\sum_{k,\tilde\sigma}\varepsilon_k n_{k,\tilde\sigma}\Big)\Big(\sum_{k'} a_{\sigma,\sigma'}(k')c^\dagger_{k',\sigma}c^\dagger_{-k',\sigma'}|\phi_0\rangle\Big) && (46)\\
&= \langle\psi_{\sigma\sigma'}|\Big(\sum_{\substack{k\le k_F\\ \tilde\sigma}}\varepsilon_k\hat n_{k,\tilde\sigma} + \sum_{\substack{k>k_F\\ \tilde\sigma}}\varepsilon_k\hat n_{k,\tilde\sigma}\Big)\Big(\sum_{k'} a_{\sigma,\sigma'}(k')c^\dagger_{k',\sigma}c^\dagger_{-k',\sigma'}|\phi_0\rangle\Big) && (47)\\
&= \langle\psi_{\sigma\sigma'}|\sum_{\substack{k\le k_F\\ \tilde\sigma}}\varepsilon_k n_{k,\tilde\sigma}|\psi_{\sigma\sigma'}\rangle + \sum_{k,k'} a_{\sigma,\sigma'}(k')\big(\varepsilon_{k'}\delta_{k,k'}\delta_{\sigma,\tilde\sigma}\\
&\qquad + \varepsilon_{k'}\delta_{k,-k'}\delta_{\sigma',\tilde\sigma}\big)c^\dagger_{k',\sigma}c^\dagger_{-k',\sigma'}|\phi_0\rangle && (48)\\
&= \Big(\langle\phi_0|\sum_k c_{-k,\sigma'}c_{k,\sigma}a^*_{\sigma,\sigma'}(k)\Big)\Big(\sum_{\substack{k\le k_F\\ \tilde\sigma}}\varepsilon_k n_{k,\tilde\sigma}|\psi_{\sigma\sigma'}\rangle\\
&\qquad + \sum_{k'} a_{\sigma,\sigma'}(k')(\varepsilon_{k'}+\varepsilon_{-k'})c^\dagger_{k',\sigma}c^\dagger_{-k',\sigma'}|\phi_0\rangle\Big) && (49)\\
&= \sum_k |a_{\sigma,\sigma'}(k)|^2\Big((\varepsilon_k+\varepsilon_{-k}) + \sum_{\substack{k\le k_F\\ \tilde\sigma}}\varepsilon_k n_{k,\tilde\sigma}\Big) && (50)\\
&= \sum_k |a_{\sigma,\sigma'}(k)|^2(\varepsilon_k+\varepsilon_{-k}) + c\sum_k |a_{\sigma\sigma'}(k)|^2, && (51)
\end{aligned}$$

where c collects the (constant) energy of the filled Fermi sea. From $\langle\psi_{\sigma\sigma'}|\psi_{\sigma\sigma'}\rangle \overset{!}{=} 1$ it follows that $\sum_k |a_{\sigma\sigma'}(k)|^2 \overset{!}{=} 1$, cf. (5.29), and therefore

$$H_0 = \sum_k |a_{\sigma,\sigma'}(k)|^2(\varepsilon_k+\varepsilon_{-k}), \qquad \Longrightarrow \qquad H = H_0 + H_T = (5.29). \tag{52}$$

The step $\varepsilon_k = \varepsilon_{-k}$ needed for (5.30) is true due to Kramers' theorem; see e.g. Sec. 5.3 of Czycholl I.

When does the gap equation degenerate?

For (5.32) it was assumed that $C \equiv \sum_k a_{\sigma\sigma'}(k) \neq 0$. Let us examine when C is actually zero. We have

$$a_{\sigma\sigma'}(k) = \frac{V\sum_{k'} a_{\sigma\sigma'}(k')}{2\varepsilon_k - \lambda}, \qquad (5.31) \tag{53}$$

where $\varepsilon_k = \varepsilon_{-k}$ by Kramers' theorem, and hence

$$a_{\sigma\sigma'}(-k) = a_{\sigma\sigma'}(k). \tag{$*$}$$

Moreover, for equal spins,

$$|\psi_{\sigma\sigma}\rangle = \sum_k a_{\sigma,\sigma}(k) c^\dagger_{k,\sigma} c^\dagger_{-k,\sigma} |\phi_0\rangle \tag{54}$$

$$= \sum_k a_{\sigma,\sigma}(k) \big(\delta_{k,-k} - c^\dagger_{-k,\sigma} c^\dagger_{k,\sigma}\big) |\phi_0\rangle \tag{55}$$

$$= -\sum_k a_{\sigma,\sigma}(k) c^\dagger_{-k,\sigma} c^\dagger_{k,\sigma} |\phi_0\rangle \tag{56}$$

$$= -\sum_k a_{\sigma,\sigma}(-k) c^\dagger_{k,\sigma} c^\dagger_{-k,\sigma} |\phi_0\rangle \tag{57}$$

$$= -\sum_k a_{\sigma,\sigma}(k) c^\dagger_{k,\sigma} c^\dagger_{-k,\sigma} |\phi_0\rangle = -|\psi_{\sigma\sigma}\rangle. \tag{58}$$

Here $\delta_{k,-k}$ drops out because $\sum_k$ does not include momenta inside the Fermi sea and thus $k \neq 0$; in the fourth line $k \to -k$ was substituted, and in the fifth line $(*)$ was used. Consequently

$$\langle\psi_{\sigma\sigma}|\psi_{\sigma\sigma}\rangle = -\langle\psi_{\sigma\sigma}|\psi_{\sigma\sigma}\rangle \implies \langle\psi_{\sigma\sigma}|\psi_{\sigma\sigma}\rangle = 0 \implies \sum_k |a_{\sigma\sigma}(k)|^2 = 0 \implies a_{\sigma\sigma}(k) = 0, \tag{59}$$

so $\sum_k a_{\sigma\sigma}(k) = 0$. Thus $\sigma = \sigma'$ yields $C = 0$, cf. (5.35), and Cooper pairs consist of electrons with opposite spins.

Note that if $\psi(r_1 - r_2)$ is antisymmetric, then $\psi(0) = 0$ and $C = 0$ as well, so the derivation of (5.35) is different in that case.

Position representation.

Project the state $|\psi_{\sigma,\sigma'}\rangle = \sum_k a_{\sigma,\sigma'}(k) c^\dagger_{k,\sigma} c^\dagger_{-k,\sigma'} |\phi_0\rangle$ onto the position basis. Since H acts trivially on the spin space ($\sigma \to \sigma$, $\sigma' \to \sigma'$), one can factorize the wave function:

$$\langle \boldsymbol{r}_1, \boldsymbol{r}_2 | \psi_{\sigma,\sigma'}\rangle = \psi_{\sigma,\sigma'}(\boldsymbol{r}_1, \boldsymbol{r}_2) = \psi(\boldsymbol{r}_1, \boldsymbol{r}_2)\, \chi(\sigma, \sigma'), \qquad (5.37) \tag{60}$$

with

$$\psi(\boldsymbol{r}_1, \boldsymbol{r}_2) = \frac{1}{V} \sum_k a(k)\, e^{i\boldsymbol{k}\cdot\boldsymbol{r}_1} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_2} = \psi(\boldsymbol{r}_1 - \boldsymbol{r}_2), \qquad (5.36) \tag{61}$$

using $\langle \boldsymbol{r}|\boldsymbol{k}\rangle = \frac{1}{\sqrt{V}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$.

1.2.4 BCS theory (Czycholl 5.4)

Mean-field recap.

Write $\hat{n}_\alpha = \langle n_\alpha\rangle + \delta\hat{n}_\alpha$ and assume that $\delta\hat{n}_\alpha$ is small, such that second-order fluctuations can be dropped:

$$\hat{n}_\alpha \hat{n}_\beta = \big(\langle\hat{n}_\alpha\rangle + \delta\hat{n}_\alpha\big)\big(\langle\hat{n}_\beta\rangle + \delta\hat{n}_\beta\big) \tag{62}$$

$$\overset{\delta\hat{n}_\alpha\delta\hat{n}_\beta \approx 0}{=} \langle\hat{n}_\alpha\rangle\langle\hat{n}_\beta\rangle + \langle\hat{n}_\alpha\rangle\big(\hat{n}_\beta - \langle\hat{n}_\beta\rangle\big) + \langle\hat{n}_\beta\rangle\big(\hat{n}_\alpha - \langle\hat{n}_\alpha\rangle\big) \tag{63}$$

$$= \langle\hat{n}_\alpha\rangle\hat{n}_\beta + \langle\hat{n}_\beta\rangle\hat{n}_\alpha - \langle\hat{n}_\alpha\rangle\langle\hat{n}_\beta\rangle. \tag{64}$$

Now apply this to $c^{(\dagger)}_{-k\downarrow}c^{(\dagger)}_{k\uparrow}$ instead of $\hat{n}_\alpha$ to derive a mean-field version of the BCS Hamiltonian, which gives (5.40).

Nambu form of the effective Hamiltonian.

Rewrite H_{eff} as (following Becca & Sorella, and the Superconductivity IMPRS UFAST 2017 course of Michael Sentef)

$$\tilde{H}_{\text{eff}} = \sum_k \begin{pmatrix} c^\dagger_{k\uparrow} & c_{-k\downarrow} \end{pmatrix} \begin{pmatrix} \varepsilon_k & -\Delta \\ -\Delta^* & -\varepsilon_k \end{pmatrix} \begin{pmatrix} c_{k\uparrow} \\ c^\dagger_{-k\downarrow} \end{pmatrix} \tag{65}$$

$$= \sum_k \begin{pmatrix} c^\dagger_{k\uparrow} & c_{-k\downarrow} \end{pmatrix} \begin{pmatrix} \varepsilon_k c_{k\uparrow} - \Delta c^\dagger_{-k\downarrow} \\ -\Delta^* c_{k\uparrow} - \varepsilon_k c^\dagger_{-k\downarrow} \end{pmatrix} \tag{66}$$

$$= \sum_k \varepsilon_k c^\dagger_{k\uparrow}c_{k\uparrow} - \Delta c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow} - \Delta^* c_{-k\downarrow}c_{k\uparrow} - \varepsilon_k \underbrace{c_{-k\downarrow}c^\dagger_{-k\downarrow}}_{=1-c^\dagger_{-k\downarrow}c_{-k\downarrow}} \tag{67}$$

$$= \sum_{k,\sigma} \varepsilon_k c^\dagger_{k\sigma}c_{k\sigma} - \Delta^* \sum_k c_{-k\downarrow}c_{k\uparrow} - \Delta \sum_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow} - \sum_k \varepsilon_k \tag{68}$$

$$= H_{\text{eff}} - \underbrace{\frac{|\Delta|^2}{V} - \sum_k \varepsilon_k}_{\text{const.}}, \qquad (5.41) \tag{69}$$

Set $M \equiv \begin{pmatrix} \varepsilon_k & -\Delta \\ -\Delta^* & -\varepsilon_k \end{pmatrix}$ and diagonalize it, such that H_{eff} becomes diagonal — this is a unitary transformation.

Diagonalizing the 2×2 matrix M.

Eigenvalues:

$$\det \begin{pmatrix} \varepsilon_k - \lambda & -\Delta \\ -\Delta^* & -\varepsilon_k - \lambda \end{pmatrix} = -(\varepsilon_k - \lambda)(\varepsilon_k + \lambda) - |\Delta|^2 \tag{70}$$

$$= -\big(\lambda^2 - \varepsilon_k^2 - |\Delta|^2\big) \stackrel{!}{=} 0 \tag{71}$$

$$\implies \quad \lambda_{1,2} = \pm\sqrt{\varepsilon_k^2 + |\Delta|^2} \equiv E_k^\pm \in \mathbb{R}. \tag{$*_1$}$$

Eigenvectors: from

$$\begin{pmatrix} \varepsilon_k - E_k^\pm & -\Delta \\ -\Delta^* & -\varepsilon_k - E_k^\pm \end{pmatrix} \begin{pmatrix} x_\pm \\ y_\pm \end{pmatrix} \stackrel{!}{=} 0 \tag{72}$$

the first row (I) gives

$$(\varepsilon_k - E_k^\pm)x_\pm - \Delta y_\pm = 0 \quad \iff \quad y_\pm = \frac{\varepsilon_k - E_k^\pm}{\Delta}\, x_\pm, \tag{I$'$}$$

while the second row (II) gives, using (I$'$) and $(E_k^\pm)^2 = \varepsilon_k^2 + |\Delta|^2$,

$$\Delta^* x_\pm + (\varepsilon_k + E_k^\pm)y_\pm \stackrel{(\text{I}')}{=} \Big(\Delta^* + \frac{\varepsilon_k^2 - (E_k^\pm)^2}{\Delta}\Big)x_\pm = 0, \tag{73}$$

i.e. it is satisfied identically. Hence $x_\pm$ can be chosen arbitrarily. $(\odot)$

Normalization.

Requiring $|x_\pm|^2 + |y_\pm|^2 \stackrel{!}{=} 1$ and using (I′),

$$|x_\pm|^2\Big(1 + \frac{(\varepsilon_k - E_k^\pm)^2}{|\Delta|^2}\Big) = |x_\pm|^2\,\frac{|\Delta|^2 + (\varepsilon_k - E_k^\pm)^2}{|\Delta|^2} \stackrel{!}{=} 1, \tag{74}$$

where

$$\begin{aligned}(\varepsilon_k - E_k^\pm)^2 &= \varepsilon_k^2 - 2\varepsilon_k E_k^\pm + (E_k^\pm)^2 &(75)\\ &= \varepsilon_k^2 - 2\varepsilon_k E_k^\pm + \varepsilon_k^2 + |\Delta|^2 &(76)\\ &= 2\varepsilon_k^2 - 2\varepsilon_k E_k^\pm + |\Delta|^2. &(77)\end{aligned}$$

Therefore

$$\begin{aligned}|x_\pm|^2 &= \frac{|\Delta|^2}{2\big(\varepsilon_k^2 - \varepsilon_k E_k^\pm + |\Delta|^2\big)} \stackrel{(*_1)}{=} \frac{1}{2}\,\frac{|\Delta|^2}{(E_k^\pm)^2 - \varepsilon_k E_k^\pm} \stackrel{(*_1)}{=} \frac{1}{2}\,\frac{(E_k^\pm)^2 - \varepsilon_k^2}{E_k^\pm\big(E_k^\pm - \varepsilon_k\big)} &(78)\\ &= \frac{1}{2}\,\frac{E_k^\pm + \varepsilon_k}{E_k^\pm} = \frac{1}{2}\Big(1 + \frac{\varepsilon_k}{E_k^\pm}\Big) = \frac{1}{2}\Big(1 \pm \frac{\varepsilon_k}{E_k}\Big), &(79)\\ |y_\pm|^2 &= 1 - |x_\pm|^2 = \frac{1}{2}\Big(1 - \frac{\varepsilon_k}{E_k^\pm}\Big) = \frac{1}{2}\Big(1 \mp \frac{\varepsilon_k}{E_k}\Big). &(\mathrm{A})\end{aligned}$$

Fixing the free phase.

Choose $x_\pm$ to be some real value, which is allowed by ($\odot$). Then (A) gives $x_\pm^2 = y_\mp^2$, i.e. $x_\pm = \pm y_\mp$, which is still too many solutions. Set

$$x_+ = y_- \equiv u_k, \qquad x_- = -y_+ \equiv v_k. \tag{80}$$

Since $M_k^\dagger = M_k$, we know that M_k can be diagonalized by a unitary U_k:

$$M = \begin{pmatrix}\varepsilon_k & -\Delta\\ -\Delta^* & -\varepsilon_k\end{pmatrix} = U_k^\dagger \begin{pmatrix}E_k^+ & 0\\ 0 & E_k^-\end{pmatrix} U_k, \qquad U_k = \begin{pmatrix}x^+ & x^-\\ y^+ & y^-\end{pmatrix} = \begin{pmatrix}u_k & v_k\\ -v_k & u_k\end{pmatrix}, \tag{81}$$

which is obviously unitary.

The Bogoliubov transformation.

Inserting this into the Hamiltonian yields

$$\begin{aligned}\tilde{H}_{\text{eff}} &= \sum_k \begin{pmatrix}c_{k\uparrow}^\dagger & c_{-k\downarrow}\end{pmatrix} U_k U_k^\dagger \begin{pmatrix}\varepsilon_k & -\Delta\\ -\Delta^* & -\varepsilon_k\end{pmatrix} U_k U_k^\dagger \begin{pmatrix}c_{k\uparrow}\\ c_{-k\downarrow}^\dagger\end{pmatrix} - \sum_k \varepsilon_k &(82)\\ &= \sum_k \begin{pmatrix}\alpha_k^\dagger & \beta_k\end{pmatrix} \begin{pmatrix}E_k^+ & 0\\ 0 & E_k^-\end{pmatrix} \begin{pmatrix}\alpha_k\\ \beta_k^\dagger\end{pmatrix} - \sum_k \varepsilon_k &(83)\\ &= \sum_k E_k^+ \alpha_k^\dagger \alpha_k + E_k^- \beta_k \beta_k^\dagger - \sum_k \varepsilon_k &(84)\\ &= \sum_k E_k^+\big(\alpha_k^\dagger \alpha_k + \beta_k^\dagger \beta_k\big) - \underbrace{\sum_k \big(\varepsilon_k + E_k^+\big)}_{\text{const.}}, \qquad \sim (5.46) &(85)\end{aligned}$$

where $\{\beta_k, \beta_k^\dagger\} = 1$ and $E_k^- = -E_k^+$ were used, and with $\begin{pmatrix}\alpha_k \\ \beta_k^\dagger\end{pmatrix} = U_k^\dagger \begin{pmatrix}c_{k\uparrow} \\ c_{-k\downarrow}^\dagger\end{pmatrix}$. Here U_k is known as the Bogoliubov transformation, with the corresponding Bogoliubov quasiparticles

$$\alpha_k = u_k c_{k\uparrow} - v_k c_{-k\downarrow}^\dagger, \tag{86}$$

$$\beta_k = u_k c_{-k\downarrow} + v_k c_{k\uparrow}^\dagger, \tag{87}$$

which reproduces (5.43)–(5.47). They obey the usual fermionic anticommutation relations, as can be seen as follows (for arbitrary unitary transformations).

Nambu spinors and the anticommutation relations.

Define

$$\psi_k = \begin{pmatrix}c_{k\uparrow} \\ c_{-k\downarrow}^\dagger\end{pmatrix}, \qquad \psi_k^\dagger = \begin{pmatrix}c_{k\uparrow}^\dagger & c_{-k\downarrow}\end{pmatrix} \qquad \text{(Nambu spinors)}, \tag{88}$$

with $\{\psi_{k,i}, \psi_{k,j}^\dagger\} = \psi_{k,i}\psi_{k,j}^\dagger + \psi_{k,j}^\dagger\psi_{k,i} = \delta_{ij}$. The Bogoliubov transformation can then be written as

$$\Gamma_k \equiv \begin{pmatrix}\alpha_k \\ \beta_k^\dagger\end{pmatrix} = U^\dagger \psi_k, \qquad \Gamma_k^\dagger \equiv \psi_k^\dagger U, \tag{89}$$

so that, using bilinearity of $\{\cdot,\cdot\}$ and that the $U_{\alpha\beta}^{(\dagger)}$ are numbers,

$$\{\Gamma_{k,i}, \Gamma_{k,j}^\dagger\} = \Big\{ \sum_\ell U_{i,\ell}^\dagger \psi_{k,\ell},\ \sum_m \psi_{k,m}^\dagger U_{m,j} \Big\} \tag{90}$$

$$= \sum_{\ell,m} U_{i,\ell}^\dagger \{\psi_{k,\ell}, \psi_{k,m}^\dagger\} U_{m,j} \tag{91}$$

$$= \sum_\ell U_{i,\ell}^\dagger U_{\ell,j} = (U^\dagger U)_{ij} = \delta_{ij}. \tag{92}$$

Since U is unitary, α_k and β_k are fermionic operators.

Coherence factors.

With (5.45),

$$\big(u_k v_k\big)^2 = \frac{1}{4}\Big(1 - \frac{\varepsilon_k^2}{(E_k^+)^2}\Big) = \frac{1}{4}\Big(\frac{\varepsilon_k^2 + |\Delta|^2 - \varepsilon_k^2}{\varepsilon_k^2 + |\Delta|^2}\Big) \tag{93}$$

$$= \frac{1}{4}\,\frac{|\Delta|^2}{(E_k^+)^2} \qquad \Longrightarrow \qquad u_k v_k = \pm\frac{1}{2}\,\frac{|\Delta|}{E_k^+}, \tag{94}$$

from which the rest follows, i.e. (5.51), where the negative solution leads to the contradiction $1 =$ negative value.

The BCS ground state.

Since α_k is fermionic, $n_{\alpha,k}^2 \equiv (\alpha_k^\dagger\alpha_k)^2 = \alpha_k^\dagger\alpha_k$, and thus any eigenvalue λ of $n_{\alpha,k}$ must satisfy $\lambda^2 = \lambda$, i.e. $\lambda \in \{0, 1\}$ (and the same for $n_{\beta,k}$). Since $E_k^+ \geq 0$, it follows that if $\alpha_k|\psi_{\rm BCS}\rangle = \beta_k|\psi_{\rm BCS}\rangle = 0$, then $|\psi_{\rm BCS}\rangle$ is the ground state of $H_{\rm eff} = \sum_k E_k^+(\alpha_k^\dagger\alpha_k + \beta_k^\dagger\beta_k)$.

The pairing ansatz.

Following Becca & Sorella (1.91),

$$|\psi_{\rm BCS}\rangle = \exp\Big(\sum_k f_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow}\Big)|0\rangle \tag{95}$$

$$= \prod_k \exp\big(f_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow}\big)|0\rangle \tag{96}$$

$$= \prod_k \Big[\sum_{n=0}^{\infty}\frac{f_k^n}{n!}\big(c^\dagger_{k,\uparrow}\big)^n\big(c^\dagger_{-k,\downarrow}\big)^n\Big]|0\rangle, \qquad \text{where } \big(c^\dagger_\alpha\big)^2|0\rangle = 0, \tag{97}$$

$$= \prod_k \big(1 + f_k c^\dagger_{k,\uparrow}c^\dagger_{-k,\downarrow}\big)|0\rangle \tag{98}$$

$$= \bigotimes_k \Big(|0\rangle_k + f_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow}|0\rangle_k\Big), \tag{99}$$

with the one-momentum vacuum $|0\rangle_k$ defined by $|0\rangle = \bigotimes_k |0\rangle_k$ and $c^\dagger_{k\uparrow}|0\rangle_k = c^\dagger_{-k\downarrow}|0\rangle_k = 0$. Thus this ansatz is a pure product state, $|\psi_{\rm BCS}\rangle = \bigotimes_k |\psi_{\rm BCS}\rangle_k$, and therefore

$$\alpha_k|\psi_{\rm BCS}\rangle = \alpha_k \bigotimes_{k'} |\psi_{\rm BCS}\rangle_{k'} = |\psi\rangle_{k_1}\otimes\cdots\otimes\alpha_k|\psi_{\rm BCS}\rangle_k\otimes\ldots \tag{100}$$

so that only a single momentum sector has to be examined:

$$\alpha_k|\psi_{\rm BCS}\rangle_k = \big(u_k c_{k\uparrow} - v_k c^\dagger_{-k\downarrow}\big)\Big(|0\rangle_k + f_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow}|0\rangle_k\Big) \tag{101}$$

$$= \big(u_k f_k c_{k\uparrow}c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow} - v_k c^\dagger_{-k\downarrow}\big)|0\rangle_k \stackrel{!}{=} 0 \tag{102}$$

$$\iff \quad v_k c^\dagger_{-k\downarrow}|0\rangle_k = u_k f_k\big(1 - c^\dagger_{k\uparrow}c_{k\uparrow}\big)c^\dagger_{-k\downarrow}|0\rangle_k \tag{103}$$

$$= u_k f_k\Big(c^\dagger_{-k\downarrow} + c^\dagger_{k\uparrow}\underbrace{c^\dagger_{-k\downarrow}c_{k\uparrow}}_{=0}\Big)|0\rangle_k, \tag{104}$$

where $c_\alpha|0\rangle = 0$ and $c^\dagger_\alpha c^\dagger_\alpha|0\rangle = 0$ were used. Since $u_k \neq 0$ this gives $f_k = v_k/u_k$, and similarly

$$\beta_k|\psi_{\rm BCS}\rangle_k = \big(u_k f_k c_{-k\downarrow}c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow} + v_k c^\dagger_{k\uparrow}\big)|0\rangle_k \stackrel{!}{=} 0 \tag{105}$$

$$\iff \quad v_k c^\dagger_{k\uparrow}|0\rangle_k = u_k f_k\, c^\dagger_{k\uparrow}\big(1 - c^\dagger_{-k\downarrow}c_{-k\downarrow}\big)|0\rangle_k, \tag{106}$$

again yielding $f_k = v_k/u_k$.

Therefore $|\psi_{\rm BCS}\rangle$ is indeed the ground state of the mean-field Hamiltonian, and it behaves like the vacuum of the Bogoliubons. To avoid problems due to $u_k = 0$, one often writes instead

$$|\psi_{\rm BCS}\rangle = \prod_k \big(u_k + v_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow}\big)|0\rangle. \tag{107}$$

1.3 References

- G. Czycholl, *Solid State Theory*, Volume 2 (and Sec. 5.3 of Volume 1 for Kramers' theorem).

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- M. Sentef, *Superconductivity*, IMPRS UFAST 2017 course.